# 1 Fundamental Theorem of Linear Maps

**Theorem 1.1 (Fundamental Theorem of Linear Maps)**

Let $T : V \to W$ be a linear map with $V$ finite-dimensional. Then $\text{Im}(T)$ is finite-dimensional and

$$\dim(V) = \dim(\ker(T)) + \dim(\text{Im}(T)). \tag{1}$$

**Exercise 1.2**

Complete the following proof of 1.1.

**Proof:**

Let $\{u_1, ..., u_n\}$ be a basis of $\ker(T)$. Then $\dim(\ker(T)) = m$. By the Plus/Minus Lemma, the linearly independent set $\{u_1, ..., u_m\}$ can be extended to a basis $\{u_1, ..., u_m, v_1, ..., v_n\}$ of $V$. Then, $\dim(V) = m + n$. To complete the proof, we need to show $\text{Im}(T)$ is finite-dimensional and $\dim(\text{Im}(T)) = n$.

(a) Show that $\{Tv_1, ..., Tv_n\}$ spans $\text{Im}(T)$.

We have that $\text{Im}(T) = \{T(v) \mid v \in V\}$. But, $v = \sum_{i=1}^{m} \alpha_i u_i + \sum_{i=1}^{n} \beta_i v_i$. But then, applying $T$ gives:

$$\begin{aligned} T(v) &= T(\sum_{i=1}^{m} \alpha_i u_i + \sum_{i=1}^{n} \beta_i v_i) \\ &= T(\sum_{i=1}^{m} \alpha_i u_i) + T(\sum_{i=1}^{n} \beta_i v_i) \\ &= 0 + T(\sum_{i=1}^{n} \beta_i v_i) \\ &= \sum_{i=1}^{n} \beta_i T(v_i), \end{aligned}$$

showing the $T(v_i)$ span $V$.

(b) Show that $\text{span}(Tv_1, ..., Tv_n\}$ is also independent and therefore a basis of $\text{Im}\{T\}$.

If $T(v_i) = T(v_j)$ for $i \neq j$, then $T(v_i - v_j) = 0$. But then, $v_i - v_j$ is a linear combination of the $u_i$ (since they are a basis for the kernel), and so $v_i = v_j + \sum_{i=1}^{m} \gamma_i u_i$, i.e $v_i$ is a linear combination of the other basis vectors, which is impossible since the $\{u_1, ..., u_m, v_1, ..., v_n\}$ are a basis. So, this never happens and therefore the $v_i$ are independent.

(c) Explain why this completes the proof.

We know that $\dim(V) = m + n$, but $\dim(\ker(T)) = m$, $\dim(\text{Im}(T)) = n$, so $\dim(V) = \dim(\ker(T)) + \dim(\text{Im}(T))$ as desired.

**Corollary 1.3**

Let $V$ and $W$ be finite-dimensional vector space, and let $T : V \to W$ be a linear map.

(a) If $\dim(V) > \dim(W)$, then $T$ is not injective.

(b) If $\dim(V) < \dim(W)$, then $T$ is not surjective.

**Proof:**

(a) We know $\dim(\text{Im}(T)) \leq \dim(W)$, so if $\dim(V) > \dim(W)$, then $\dim(V) > \dim(\text{Im}(T))$. But, since $\dim(V) = \dim(\ker(T)) + \dim(\text{Im}(T))$, we see $\dim(\ker(T)) + \dim(\text{Im}(T)) > \dim(\text{Im}(T))$, so therefore $\dim(\ker(T)) > 0$.

(b) We know $\dim(\ker(T))+\dim((\text{Im}(T)) = \dim(V)$, and since $\dim(\ker(T)) \geq 0$, we see $\dim(\text{Im}(T)) \leq \dim(V) < \dim(W)$. But, if $\dim((T)) < \dim(W)$, $(\dim((T))$ cannot be all of $W$ (otherwise it would have $\dim(W)$) and therefore the map is not surjective.

**Exercise 1.4**

Use Theorem 1.1 to prove Corollary 1.4.

**Exercise 1.5**

Consider the linear map $T : F^4 \to F^3$ defined by

$$T(z_1, z_2, z_3, z_4) = (\sqrt{7}z_2 + \pi z_2 + z_4, 97z_1 + 3z_2 + 2z_3, z_2 + 6z_3 + 7z_4).$$

Prove that $T$ is not injective.

**Solution 1**

Since $\dim(F^4) = 4 > \dim(F^3) = 3$, by the corollary the map cannot be injective.

## 2 Systems of Linear Equations, Revisted

In the first unit, we represented a system of linear equations

$$\begin{array}{ccccccccc} A_{11}x & + & A_{12}x_2 & + & \ldots & + & A_{1n}x_n & = & y_1 \\ & A_{21}x & + & A_{22}x_2 + & \ldots + & A_{2n}x_n & = & y_1 & \\ \vdots & & \vdots & & \vdots & & \vdots & \vdots & \\ A_{m1}x & + & A_{m2}x_2 & + & \ldots & + & A_{mn}x_{mn} & = & y_m \end{array} \tag{2}$$

as a matrix equation

$$Ax = y. \tag{3}$$

As we saw last time, we can interpret (2.2) as saying that $y \in M^{m\times 1}(F)$ is in the image of the linear map $T : M^{n\times 1}(F) \to M^{m\times 1}(F)$ which multiplies $x$ by the coefficient matrix $A : T(x) = Ax$.

**Exercise 2.1**

Use Theorem 1.1 to prove the following:

(a) a homogenous system of linear equations with more variables than equations has nonzero solutions.

(b) A system of linear equations with more equations than variables has no solution for some choice of constant terms.



**Proof:**

Let $m$ be the number of equations and $m$ the number of variables. Then, this matrix is a map from $M^{n\times 1}(F) \to M^{m\times 1}(F)$, which are of dim $n$ and dim $m$ respectively.

In the homogenous case, this is saying $n > m$, in which case by the corollary, this is not injective, meaning the kernel is not trivial, meaning there is some nontrivial $v$ such that $Av = 0$, i.e a nontrivial solution of the system.

**(b)** This is saying $n < m$, in which case the map is not surjective, meaning for some $(y_1, ..., y_m)$, there is no $(x_1, ..., x_n)$ that maps to it, i.e it has no solution.





**Exercise 2.2**

Without doing any calculations, determine whether the following system is consistent for all $b_1, b_2, b_3$.

$$\begin{aligned} 3x_1 + 2x_2 &= b_1 \\ -2x_1 + 4x_3 &= b_2 \\ x_1 + 5x_2 &= b_3. \end{aligned}$$



**Solution 2**

Since we have more equations than variables, for some $b_1, b_2, b_3$ there is no solution for some $b_1, b_2, b_3$.

## 3 Exercises



**Problem 1**

Suppose $T \in \mathcal{L}(F^4, F^2)$ such that $\ker(T) = \{(x_1, x_2, x_3, x_4) \in F^4 : x_1 = 5x_2 \text{ and } x_3 = 7x_4\}$. Prove that $T$ is surjective.



**Proof:**

The kernel has a basis by $(5, 1, 0, 0)$ and $(0, 0, 7, 0)$, so it is of dimension 2. Therefore, $\dim(V) = 4 = \dim(\ker) + \dim(\text{Im}) = 2 + \dim(\text{Im})$, so $\dim(\text{Im}) = 2 = \dim(F^2)$, so $\text{Im} = F^2$ and the map is surjective.





**Problem 2**

Prove that there does not exist $T \in \mathcal{L}(\mathbf{R}^5)$ such that $\text{Im}(T) = \ker(T)$.



**Proof:**

If $\text{Im}(T) = \ker(T)$, then $\dim(\text{Im}(T)) = \dim(\ker(T))$, so $\dim(\mathbf{R}^5) = 5 = 2\dim(\ker(T))$, but $\dim(\ker(T))$ is an integer, so this cannot happen.





**Problem 3**

Suppose $T \in \mathcal{L}(V, W)$ is injective and $\{v_1, ..., v_n\}$ is a linearly independent subset of $V$. Prove that $\{Tv_1, ..., Tv_n\}$ is a linearly independent subset of $W$.



**Proof:**

If $\sum_{i=1}^{n} \alpha_i T(v_i) = 0$, by linearity, $T(\sum_{i=1}^{n} \alpha_i v_i) = 0$. But then, since this is injective, the kernel is trivial, so $\sum_{i=1}^{n} \alpha_i v_i = 0$, but then since the $v_i$ are independent in $v_i$, all the $\alpha_i$ are zero. This shows independence of the $Tv_i$.





**Problem 4**

Show that if $\{v_1, ..., v_n\}$ spans $V$ and $T \in \mathcal{L}(V, W)$, then $\{Tv_1, ..., Tv_n\}$ spans $\text{Im}(T)$.



**Proof:**

If $\{v_1, ..., v_n\}$ spans $V$, then for every $v \in V$, $v = \sum_{i=1}^{n} \alpha_i v_i$, so $T(v) = T(\sum_{i=1}^{n} \alpha_i v_i) = \sum_{i=1}^{n} \alpha_i T(v_i)$, and since $\text{Im}(T)$ is exactly the set of $T(v)$, this shows that the $T(v_i)$ span $\text{Im}(T)$.





**Problem 5**

Show that for $W = \{T \in \mathcal{L}(\mathbf{R}^5, \mathbf{R}^4) \mid \dim(\ker(T)) > 2\}$ is not a subspace of $\mathcal{L}(\mathbf{R}^5, \mathbf{R}^4)$.



**Proof:**

Consider the map $T$ sending $(a, b, c, d, e)$ to $(0, 0, 0, d, e)$, and the map $S$ sending $(a, b, c, d, e)$ to $(a, b, 0, 0, 0)$. These both have kernel of dimension 3, but $S + T$ sends $(a, b, c, d, e)$ to $(a, b, 0, d, e)$, which has kernel of dimension $1 < 2$, so this is not in $W$. Therefore, $W$ is not closed under addition and cannot be a subspace.

